

Ch35: 28, 41, 42, 47, 56, 81

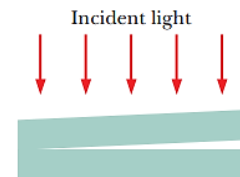
Basic score 4' +problem 41 3'+ problem 56 3'

28. There is a phase shift when the light is incident from the air film to the bottom glass and reflected by the air-glass interface, so the interference maximum is given by:

$$2d = (m+1/2) \lambda_{\text{air}} \quad m=0,1,2 \quad \text{bright}$$

$$2d = (m) \lambda_{\text{air}} \quad m=0,1,2 \quad \text{dark}$$

Where d is the thickness of the air film



(a) The third sentence in the problem (Initially, a dark fringe lies at the left end, a bright fringe lies at the right end, and nine dark fringes lie between those two end fringes) implies:

$$2d_0 = m_0 \lambda_{\text{air}} = (9 + 1/2) \lambda_{\text{air}} = 9.5 \lambda_{\text{air}}$$

Then, $\Delta t = 15$ s later, we have $m' = 9.0$ in $2d' = m' \lambda_{\text{air}}$. This means

$$|\Delta d| = d_0 - d' = \frac{1}{2} (m_0 \lambda_{\text{air}} - m' \lambda_{\text{air}}) = \frac{1}{2} (9.5 - 9) \lambda_{\text{air}} = 157.5 \text{ nm}.$$

Thus, $|\Delta d|$ divided by Δt gives 10.5 nm/s.

(b) In this case, $m_f = 6$ in the formula $2d = m \lambda_{\text{air}}$, so that

$$d_0 - d_f = \frac{1}{2} (9.5 \lambda_{\text{air}} - 6 \lambda_{\text{air}}) = \frac{7}{4} \lambda_{\text{air}} = 1102.5 \text{ nm} = 1.10 \text{ } \mu\text{m}.$$

41. The maxima of a two-slit interference pattern are at angles θ given by: $d \sin \theta = m\lambda$, where d is the slit separation, λ is the wavelength, and m is an integer. So

$$d \sin \theta_1 = m\lambda_1 = 3\lambda_1 \Rightarrow \sin \theta_1 = 3\lambda_1 / d$$

$$d \sin \theta_2 = m\lambda_2 = 3\lambda_2 \Rightarrow \sin \theta_2 = 3\lambda_2 / d$$

1 point

$$\tan \theta = \frac{y}{D} \Rightarrow y = D \tan \theta$$

1 point

In this problem, the angle is small enough, so we can use small angle approximation: $\sin \theta \approx \theta \approx \tan \theta$, so we have,

$$\Delta y = D(\tan \theta_2 - \tan \theta_1) \approx D(\sin \theta_2 - \sin \theta_1) = \left[\frac{3D}{d} \right] (\lambda_2 - \lambda_1) \quad 0.5 \text{ point}$$

$$= \left[\frac{3(1.8 \text{ m})}{5.0 \times 10^{-3} \text{ m}} \right] (600 \times 10^{-9} \text{ m} - 480 \times 10^{-9} \text{ m}) = 1.44 \times 10^{-3} \text{ m} \quad 0.5 \text{ point}$$

42. (a) The exiting angle is 55° ; the same as the incident angle, due to what one might call the “transitive” nature of Snell’s law: $n_{\text{air}} \sin \theta_1 = n_1 \sin \theta_1' = n_2 \sin \theta_2 = n_3 \sin \theta_3 = \dots = n_{\text{air}} \sin \theta_{\text{exiting}}$

(b) Due to the fact that the speed (in a certain medium) is c/n (where n is that medium’s index of refraction) and that speed is distance divided by time (while it’s constant), we find

$$t = \frac{s_3}{v_3} = \frac{s_3}{c/n_3} = \frac{n_3}{c} s_3 = \frac{n_3}{c} \frac{L_3}{\cos \theta_3} = \frac{n_3}{c} \frac{L_3}{\sqrt{1 - \left(\frac{\sin \theta_1}{n_3}\right)^2}} = \frac{1.45}{3 \times 10^8} \frac{25 \times 10^{-6}}{\sqrt{1 - \left(\frac{\sin 55^\circ}{1.45}\right)^2}} m = 1.47 \times 10^{-13} m$$

47. The index of refraction of oil is greater than that of the air, but smaller than that of the water. Let the indices of refraction of the air, oil, and water be n_1 , n_2 , and n_3 , respectively. Since $n_1 < n_2$ and $n_2 < n_3$, there is a phase change of π rad from both surfaces. Since the second wave travels an additional distance of $2L$, the phase difference is

$$\phi = \frac{2\pi}{\lambda_2} (2L)$$

where $\lambda_2 = \lambda / n_2$ is the wavelength in the oil. The condition for constructive interference is

$$\frac{2\pi}{\lambda_2} (2L) = 2m\pi,$$

or

$$2L = m \frac{\lambda}{n_2}, \quad m = 0, 1, 2, \dots$$

(a) For $m = 1, 2, \dots$, maximum reflection occurs for wavelengths

$$\lambda = \frac{2n_2 L}{m} = \frac{2(1.20)(380 \text{ nm})}{m} = 912 \text{ nm}, 456 \text{ nm}, 304 \text{ nm}, \dots$$

We note that only the 456 wavelength falls within the visible light range.

(b) Maximum transmission into the water occurs for wavelengths for which reflection is a minimum. The condition for such constructive interference is given by

$$2L = \left(m + \frac{1}{2}\right) \frac{\lambda}{n_2} \Rightarrow \lambda = \frac{4n_2 L}{2m + 1}$$

which yields $\lambda = 1824 \text{ nm}$, 608 nm , $364.8 \text{ nm} \dots$ for the different values of m . We note that only the 608 nm wavelength (blue) is in the visible range.

Note: A light ray reflected by a material changes phase by $\pi \text{ rad}$ (or 180°) if the refractive index of the material is greater than that of the medium in which the light is traveling. Otherwise, there is no phase change. Note that refraction at an interface does not cause a phase shift.

56. (a) Referring to Figure 35-10(a) makes clear that

$$\theta = \tan^{-1}(y/D) = \tan^{-1}(0.205/4) = 2.93^\circ$$

$$d \sin \theta = m\lambda \rightarrow m = d \sin(2.93^\circ) / \lambda = 0.354$$

Which means it is between the central maximum ($m=0$) and the first minimum ($m=0.5$). (0.5 分)

Note that the above computation could have been simplified somewhat by avoiding the explicit use of the tangent and sine functions and making use of the small-angle approximation ($\tan \theta \approx \sin \theta$).

(b) The phase difference at point P is $\phi = 2\pi d \sin \theta / \lambda = (0.354)(2\pi) = 2.224 \text{ rad}$, From Eq. 35-22, we get phase difference 0.5 分 (只有公式分)

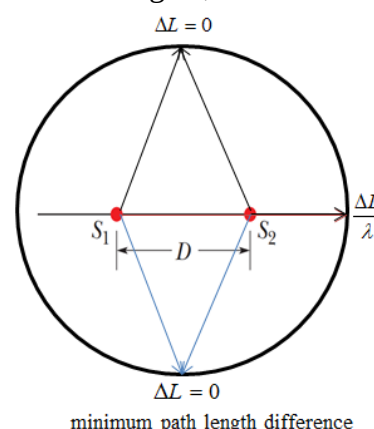
$$I = 4I_0 \cos^2(\phi/2) = 0.785 I_0 \quad 1 \text{ 分}$$

at point P and

$$I_{\text{center}} = 4I_0 \cos^2(0) = 4 I_0 \quad 0.5 \text{ 分}$$

at the center. Thus, $I / I_{\text{center}} = 0.785 / 4 = 0.196$. 0.5 分

81. The problem is asking at how many angles will there be maximum signal, and at how many will there be minimum signal? And the maxima or minima depend on the path length difference between the two point sources. The path length difference will get maximum value on the points along the two point sourced and get the minimum value 0 at the points shown in the figure. From the follow figure, we know that the path length difference changes from 0 to $(d/\lambda)\lambda = (1.8/0.5)\lambda = 3.6 \lambda$.



a) For the maxima (in phase), the path length difference is the integer number of the wavelength, that means the $\Delta L/\lambda$ can be: 0,1,2,3, 3,2,1,0,-1,-2,-3, -3,-2,-1, so there are 14 maxima

(b) The discussion about the minima directions was started in part (a). The number of values $\frac{\Delta L}{\lambda}$ in the list: 0.5, 1.5, 2.5, 3.5, 3.5, 2.5, 1.5, 0.5, -0.5, -1.5, -2.5, -3.5, -3.5, -2.5, -1.5, -0.5. There are 16 minima directions.